

An inapplicable thin-interface correction inflates the sublimation kinetic coefficient

Diagnosis, analysis and proposed correction

Working note — lunar_regolith_DSM, enceladus_DSM

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Abstract. The kinetic coefficient realized by the solver exceeds the requested `-beta_sub0` by a factor of 1.22, so that sublimation and deposition proceed 18 % more slowly than intended. The capillary length is realized correctly. The excess corresponds exactly to the two thin-interface corrections added to τ_{sub} at `lunar_main.c:829` in anticipation of their removal by the sharp-interface asymptotics; measurement indicates that no such removal occurs. An inner-region analysis shows why: for a one-sided vapor diffusivity the inner deviation of σ vanishes identically, so the vapor correction cannot arise. The thermal correction is separately misspecified, omitting the Clausius–Clapeyron coupling by which temperature acts on σ . A four-arm parameter sweep in β_{sub0} confirms the diagnosis over a range of 40: the measured ratio varies from 0.32 to 0.95 as predicted, with $\beta_{\text{fit}}/\beta_{\text{bare}} = 1.0009 \pm 0.0007$. The correction required is the removal of the two terms from τ_{sub} ; the Gibbs–Thomson relation used for comparison is correct as written.

1. Measurement

The wedge geometry places an apex-centered ice annulus in a tapering pore, with the virtual apex at $y = L_y/2$. On the centerline the interface normal is therefore exactly $\pm \hat{x}$ and v_n reduces to a one-dimensional displacement rate. `postprocess/wedge_gt_velocity.py` measures v_n on both menisci of all nine runs and compares it with

$$v_n = \frac{\sigma \rho_{vs}(T)/\rho_{vs}(T_0) - d_{0,\text{sub0}} \chi}{\beta_{\text{sub0}}}, \quad \sigma = \frac{\rho_v - \rho_{vs}(T)}{\rho_{vs}(T)}. \quad (1)$$

The solver recovers 0.826 (inner meniscus) and 0.811 (outer) of the predicted rate. Both values are constant over 90 days and across nine vapor boundary conditions whose residual driving force $\sigma - d_{0,\text{sub0}}\chi$ spans two orders of magnitude. Interface-extraction noise is 0.5 %, against an effect of 18 %, and three independent estimators of v_n agree on the median to within 1 %; the shortfall is therefore not an artifact of the measurement.

2. Identification of the affected coefficient

Rather than assume Eq. (1) and report a ratio, both coefficients may be fitted. Each snapshot supplies a triple (σ, v_n, χ) , so that

$$\sigma = \beta v_n + d_0 \chi \quad (\text{no intercept}) \quad (2)$$

constitutes a two-column least-squares problem. The design is identifiable because the two menisci carry curvature of opposite sign, the inner being concave and the outer convex, while the 3×3 reservoir grid varies v_n at nearly fixed χ :

χ	$-5.259 \times 10^3 \dots +3.459 \times 10^3 \text{ m}^{-1}$
v_n	$-1.459 \times 10^{-12} \dots +1.378 \times 10^{-12} \text{ m s}^{-1}$
$\text{corr}(v_n, \chi)$	-0.7095
condition number	2.58

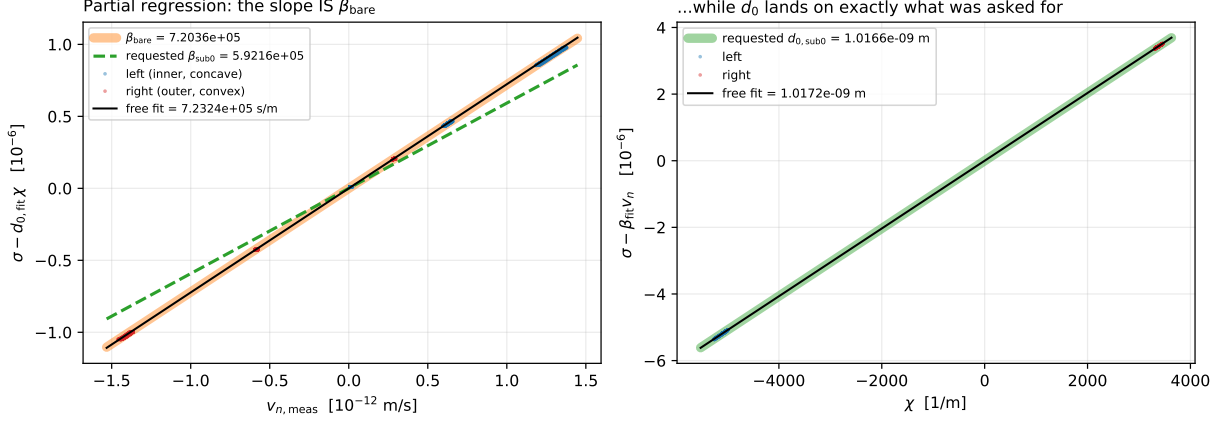


Figure 1: Partial-regression views of the fit. Left: with the fitted curvature term removed, $\sigma - d_0 \chi$ against v_n is linear through the origin with slope equal to the realized β . The fit lies within the β_{bare} band and is steeper than the requested β_{sub0} . Right: with the kinetic term removed, the slope against χ lies within the $d_{0,\text{sub0}}$ band. Both menisci, all nine runs, every fifth snapshot.

Fitting $n = 3492$ points (nine runs, two menisci, every fifth snapshot after the initial transient):

	β [s m^{-1}]	$/ \beta_{\text{sub0}}$	d_0 [m]	$/ d_{0,\text{sub0}}$
pooled	$7.2324 \times 10^5 \pm 1.8 \times 10^2$	1.2214	$1.0172 \times 10^{-9} \pm 3.5 \times 10^{-14}$	1.0006
inner	$7.1384 \times 10^5 \pm 1.6 \times 10^2$	1.2055	$1.0166 \times 10^{-9} \pm 2.7 \times 10^{-14}$	1.0000
outer	$7.2716 \times 10^5 \pm 1.3 \times 10^2$	1.2280	$1.0167 \times 10^{-9} \pm 3.5 \times 10^{-14}$	1.0001

The pooled fit gives $R^2 = 0.999997$ with an rms residual of 5.4×10^{-9} , or 0.14 % of $|\sigma|$. Holding $d_{0,\text{sub0}}$ fixed and fitting β alone gives 7.2106×10^5 , within 0.3 % of the free fit, indicating that the two parameters are not compensating one another.

Leave-one-run-out refitting displaces β over $7.2204\text{--}7.2419 \times 10^5$, a spread of 0.30 % about the mean, so no single vapor boundary condition determines the result. The residual difference between the two menisci (1.206 against 1.228) is $\pm 1\%$ and is consistent with the reported σ -extraction sensitivity of 0.7 % and 0.3 % respectively.

The capillary length is thus recovered to 0.06 %: the curvature terms of the Gibbs–Thomson relation are realized correctly, and the discrepancy resides entirely in β , as a multiplicative factor of 1.22.

3. The realized kinetic coefficient

Dividing the phase-field residual by $3 \text{ mob}/\epsilon$ gives the Moure & Fu form

$$\tau_{\text{sub}} \partial_t \phi = \epsilon^2 \nabla^2 \phi - f'(\phi) + \lambda_{\text{sub}} g(\phi) \frac{\rho_{vs}(T)}{\rho_{\text{ice}}} \sigma, \quad (3)$$

whose sharp-interface limit is Eq. (1) with realized coefficients

$$d_{0,\text{realized}} = d_{0,\text{sub0}}, \quad \beta_{\text{realized}} = \underbrace{\frac{\tau_{\text{sub}} d_{0,\text{sub0}}}{\epsilon^2}}_{\beta_{\text{bare}}} - C_{\text{corr}}. \quad (4)$$

Here C_{corr} is the $O(\epsilon)$ term generated by the variation of σ within the diffuse interface. The solver inflates τ_{sub} by the anticipated magnitude of C_{corr} , so that the two cancel and $\beta_{\text{realized}} = \beta_{\text{sub0}}$:

$$\tau_{\text{sub}} = \epsilon \lambda_{\text{sub}} \left(\underbrace{\frac{\beta_{\text{sub}}}{a_1}}_{82.2\%} + \underbrace{\frac{a_2 \epsilon}{\text{diff_sub}}}_{13.1\%} + \underbrace{\frac{a_2 \epsilon}{\text{dif_vap}}}_{4.7\%} \right), \quad \beta_{\text{bare}} = 7.2036 \times 10^5 \text{ s m}^{-1}. \quad (5)$$

The bracket is written in the $\rho_{vs}/\rho_{\text{ice}}$ -scaled variables used internally by the solver; β_{bare} is quoted in the unscaled units of β_{sub0} . The fit gives $\beta/\beta_{\text{bare}} = 1.0040$, so that $C_{\text{corr}} \approx 0$: the anticipated cancellation does not occur and the inflation is retained in full. This gives $\beta_{\text{bare}}/\beta_{\text{sub0}} = 1.2165$ and a predicted ratio of 0.8220, against measured values of 0.826 and 0.811.

The defect therefore lies in the definition of a model parameter. Eq. (1) is the intended physics and is correct; the solver does not realize the value of β supplied to it.

4. The coefficient offset Δ

4.1 Definition

Let Δ denote the amount by which the realized kinetic coefficient exceeds the requested one. From Eq. (5) with $C_{\text{corr}} = 0$,

$$\Delta \equiv \beta_{\text{bare}} - \beta_{\text{sub0}} = a_1 a_2 \epsilon \left(\frac{1}{\text{diff_sub}} + \frac{1}{\text{dif_vap}} \right) \frac{\rho_{\text{ice}}}{\rho_{vs}(T)}, \quad (6)$$

expressed in the unscaled units of β_{sub0} (s m^{-1}). The bracket is the pair of a_2 terms carried by τ_{sub} ; the factor a_1 converts the bracket into a kinetic coefficient, and $\rho_{\text{ice}}/\rho_{vs}$ removes the scaling used internally by the solver. For the wedge configuration ($\epsilon = 8.584 \times 10^{-7} \text{ m}$, $T = -20^\circ \text{C}$):

	value [s m^{-1}]	share of Δ
thermal, $a_1 a_2 \epsilon \rho_{\text{ice}}/(\rho_{vs} \text{diff_sub})$	9.44×10^4	73.7 %
vapor, $a_1 a_2 \epsilon \rho_{\text{ice}}/(\rho_{vs} \text{dif_vap})$	3.37×10^4	26.3 %
Δ	1.282×10^5	
β_{sub0} (requested)	5.9216×10^5	
$\beta_{\text{bare}} = \beta_{\text{sub0}} + \Delta$ (realized)	7.2036×10^5	

4.2 Consequences

Three properties of Eq. (6) determine its practical significance.

(i) *The offset is additive.* The resulting shortfall is $\beta_{\text{sub0}}/(\beta_{\text{sub0}} + \Delta)$ and therefore depends on β_{sub0} itself; it is not a fixed multiplicative factor. The parameter sweep of §6 exhibits this directly, the ratio varying from 0.32 to 0.95 across four values of β_{sub0} . Consequently, any correction of the form “rescale β by the observed ratio” is calibrated at a single parameter point and is inaccurate elsewhere (§7).

(ii) *The offset scales as ϵ and as $1/\rho_{vs}(T)$.* Coarser interfaces and lower temperatures both increase it, and neither quantity is freely adjustable: ϵ is fixed by resolution requirements and T by the problem under study. For the Molaro configuration ($\epsilon = 1.2 \times 10^{-7} \text{ m}$, $\alpha_c = 0.1$), $\Delta/\beta_{\text{sub0}} = 0.23$; for the wedge configuration, 0.22; at -60°C , 20.7.

(iii) *The offset imposes a lower bound on the realizable kinetic coefficient.* Since $\beta_{\text{realized}} = \beta_{\text{sub0}} + \Delta \geq \Delta$ for any positive β_{sub0} , and $\beta \propto 1/\alpha_c$, there exists an upper bound on the attachment coefficient that no choice of input can exceed:

T [°C]	ρ_{vs} [kg m ⁻³]	Δ [s m ⁻¹]	max. attainable α_c
-60	8.885×10^{-6}	1.225×10^7	0.0006
-40	1.055×10^{-4}	1.031×10^6	0.0077
-20	8.487×10^{-4}	1.282×10^5	0.0619
0	5.047×10^{-3}	2.156×10^4	0.3684

(evaluated at $\epsilon = 8.584 \times 10^{-7}$ m; halving ϵ halves Δ and doubles each bound). The Molaro configuration is affected directly: a requested $\alpha_c = 0.1$ is realized as 0.081, and in the limit $\alpha_c \rightarrow 1$, appropriate to an attachment-limited regime, Δ exceeds $\beta_{\text{sub}0}$ by a factor of 2.3.

5. Vanishing of the correction for a one-sided diffusion field

5.1 Nature of the correction

The thin-interface correction is not a physical effect. At finite ϵ the diffuse-interface formulation introduces a spurious contribution to the sharp-interface kinetic coefficient, arising because σ varies across the interfacial region in which the phase-change source is distributed. τ_{sub} is inflated by the anticipated magnitude of that contribution so that the two cancel and the model realizes $\beta_{\text{sub}0}$. The construction is standard; the question addressed here is whether the present model in fact generates the contribution for which it is being compensated.

5.2 The solvability condition

Introduce the signed normal distance ζ (positive into the vapor) and the stretched inner coordinate $\eta = \zeta/\epsilon$. In the inner region the ice phase field, written ϕ here and ϕ_i elsewhere in the project, is expanded as

$$\phi = \phi_0(\eta) + \epsilon \phi_1(\eta, s, t) + O(\epsilon^2), \quad (7)$$

in which the subscript denotes order in ϵ and is not a phase index. The leading term satisfies $\partial_{\eta\eta}^2 \phi_0 - f'(\phi_0) = 0$ and is therefore the planar, stationary equilibrium profile $\phi_0 = \frac{1}{2}(1 + \tanh(\eta/2))$; curvature, interface motion and the driving field enter at $O(\epsilon)$ through ϕ_1 .

At $O(\epsilon)$ the linearized operator is singular, and the Fredholm alternative, obtained by projection onto the translation mode $\partial_\eta \phi_0$, yields the interface condition. Writing $\sigma = \sigma_{\text{int}} + \delta\sigma(\eta)$, where σ_{int} is the outer solution extrapolated to $\zeta = 0$, the coefficients follow from

$$\int (\partial_\eta \phi_0)^2 d\eta = \frac{1}{6}, \quad \int g(\phi_0) \partial_\eta \phi_0 d\eta = \frac{1}{30}, \quad a_1 = \frac{1/6}{1/30} = 5, \quad (8)$$

evaluated numerically to six significant figures and equal to the value of a_1 used by the solver. The correction depends on a third integral,

$$I = \int g(\phi_0) \partial_\eta \phi_0 \delta\sigma(\eta) d\eta, \quad (9)$$

and vanishes if and only if I does.

The weight $g(\phi_0) \partial_\eta \phi_0$ is even in η , since $\phi_0(-\eta) = 1 - \phi_0(\eta)$ renders $g = \phi_0^2(1 - \phi_0)^2$ even and $\partial_\eta \phi_0 = \frac{1}{4} \text{sech}^2(\eta/2)$ even. The integral I therefore samples only the even part of $\delta\sigma$.

5.3 Symmetric diffusivity

For D equal in both phases and the flux partitioned as $\pm j/2$, the quasi-static inner balance $\partial_{\eta\eta}^2 \rho_v \propto \partial_\eta \phi_0$ integrates to $\partial_\eta \rho_v \propto (\phi_0 - \frac{1}{2})$, which is odd. Hence $\delta\sigma \propto \Psi(\eta) = \ln \cosh(\eta/2) - |\eta|/2 + \ln 2$ is even and decays to zero in both limits. An even deviation against an even weight gives $I \neq 0$; this is the origin of the term $-a_1 a_2 \epsilon/D$ of Karma & Rappel (1998).

5.4 One-sided diffusivity

The vapor equation of the present model carries the diffusivity $D_v \phi_a$, which vanishes in the ice. Neglecting storage, which is justified by $v_n \epsilon / (\xi_v D_v) = 2.4 \times 10^{-11}$, the inner balance is

$$\partial_\eta (D_v \phi_a \partial_\eta \rho_v) = -\epsilon v_n \rho_{\text{ice}} \partial_\eta \phi_0. \quad (10)$$

A single integration, with the constant fixed by the absence of flux into the ice at $C = \epsilon v_n \rho_{\text{ice}}$, gives

$$D_v \phi_a \partial_\eta \rho_v = \epsilon v_n \rho_{\text{ice}} (1 - \phi_0) = \epsilon v_n \rho_{\text{ice}} \phi_a \implies \partial_\eta \rho_v = \frac{\epsilon v_n \rho_{\text{ice}}}{D_v} \quad \text{wherever } \phi_a \neq 0. \quad (11)$$

The factor ϕ_a cancels, and the resulting constant is precisely the outer Stefan gradient, since $D_v \partial_\zeta \rho_v|_{\text{int}} = v_n \rho_{\text{ice}}$ yields the same value. The inner solution is thus the continuation of the outer solution through the interfacial region, so that

$$\delta\sigma \equiv 0 \implies I = 0 \implies C_{\text{corr}} = 0. \quad (12)$$

The integrand vanishes identically rather than by cancellation of contributions: since $\delta\sigma$ is null rather than merely odd, $I = 0$ for any weight, and no evaluation of the prefactor is required. Deep within the ice the floor `air_lim` = 10^{-6} on ϕ_a breaks the cancellation, but $g(\phi_0)$ there is of order 10^{-12} and the contribution is negligible.

5.5 Limitations of the present derivation

The vanishing result above is insensitive to prefactors and is correspondingly robust. The symmetric branch is less well established here: constructing $\delta\sigma = \Psi$ as above and evaluating Eq. (9) gives $a_2 = -6I = 0.087$, against the accepted value 0.158. The present treatment therefore omits $O(\epsilon)$ contributions, plausibly the curvature term, the back-reaction of ϕ_1 , or the matching of the flux partition. It is not offered as a derivation of a_2 . What is established is the structural contrast between an even, non-vanishing deviation and an identically null one; the magnitude is supplied by measurement.

5.6 The thermal channel

The thermal term is not affected by the one-sidedness argument: k is interpolated and non-zero in both phases, so a correction does exist for that channel. It is nonetheless misspecified, because `a2*eps/diff_sub` presumes that latent heat drives σ at unit strength. Temperature influences σ only through $\rho_{vs}(T)$, so the channel carries a Clausius–Clapeyron factor. Evaluating $d\sigma/d(v_n \epsilon)$ for each channel:

channel	coefficient	value
vapor	$1/D_v$	5.27×10^4
	as coded: <code>1/dif_vap</code>	4.59×10^4 (correct form; $I = 0$)
thermal	$(d\rho_{vs}/dT) L_{\text{sub}}/k$	1.0×10^2 (k_{ice}) – 1.2×10^4 (k_{air})
	as coded: <code>1/diff_sub</code>	1.29×10^5 (over-weighted by 11–1300)

Since `ThermalCond` interpolates between the two conductivities, any admissible choice renders this term negligible. This misspecification is independent of the preceding argument.

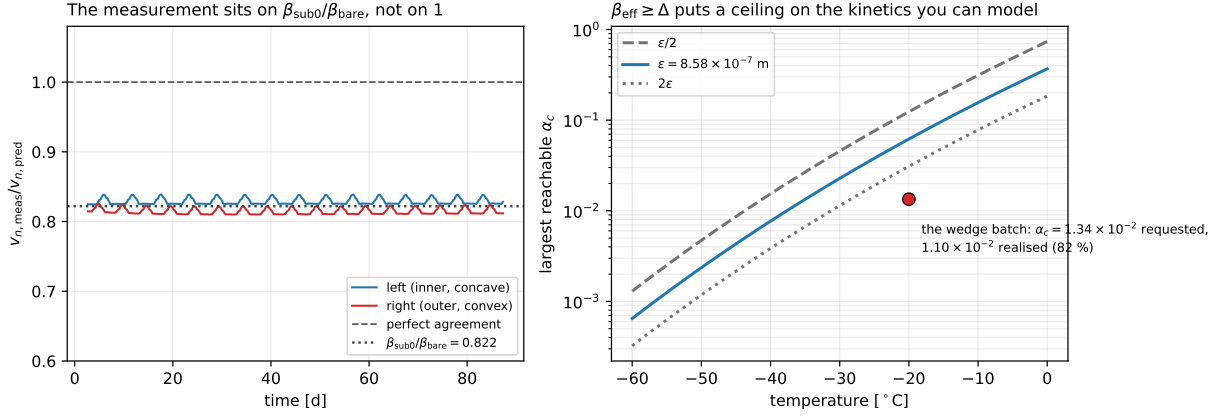


Figure 2: Left: the measured ratio follows $\beta_{\text{sub0}}/\beta_{\text{bare}}$ throughout the run on both menisci. Right: the upper bound on the attachment coefficient implied by $\beta_{\text{realized}} \geq \Delta$, as a function of temperature and ϵ .

5.7 Scope of the measurement

spurious term	form	constrained by the sweep?
kinetic correction	$\propto v_n$	yes; $\leq 0.1\%$ of β_{bare}
solute trapping	$\propto \epsilon v_n$ (shifts σ_{int})	yes; also $\propto v_n$, absorbed into fitted β
surface diffusion	tangential flux along the interface	no; distinct operator

The first two are excluded. The third is not observable in a centerline v_n measurement and is not constrained by any result reported here. The surface excess of the diffusivity weight, $\int [(1 - \phi_0) - \Theta(-\eta)] d\eta$, vanishes exactly by the same antisymmetry, which suggests that the spurious surface diffusion is likewise suppressed; this is an indication rather than a calculation. The point is material because the model contains no surface-diffusion term by construction, so a spurious one would not be evident from the source, and the Molaro analysis separately invokes physical surface diffusion. The two must be distinguished.

5.8 Alternative explanations considered

- *Grid pinning or lattice friction.* The discrepancy is a constant multiplicative factor, identical on menisci whose velocities differ by a factor of 60. A threshold or friction mechanism would not produce this behavior.
- *Bias in the interface extraction.* Three independent estimators of v_n agree on the median to within 1%; the residual scatter is 0.5%, zero-mean, and correlated with sub-cell phase.
- *Incorrect conversion of a_1 or a_2 .* The ϵ used here is Moure & Fu's ϵ , and is deliberately *not* Kaempfer & Plapp's W ($W_{\text{KP}} = \sqrt{2}\epsilon$); the constants already absorb the difference. Mapping to K&P's $[-1, 1]$ form gives $d_{0,\text{sub}} = a_1^{\text{KP}} \sqrt{2} \cdot 4 \cdot \epsilon / \lambda_{\text{sub}} = 5.0000 \epsilon / \lambda_{\text{sub}}$ against the coded $a_1 = 5.0$, and $a_1^{\text{KP}} a_2^{\text{KP}} \sqrt{2} = 0.78338$ against the coded $a_1 a_2 = 0.7905$ (0.91%). This is already documented at `enceladus_DSM/src/assembly.c:46-58`. Recovery of $d_{0,\text{sub0}}$ to 0.06% by the free fit constitutes an independent measurement of a_1 .
- *The temporal scaling ξ_v .* It multiplies the vapor diffusion and the ice source identically, so it cancels between them in the inner balance (§5) and survives only in storage/diffusion, $v_n \epsilon / (\xi_v D_v) = 2.4 \times 10^{-11}$. The unscaled `dif_vap` in Eq. (5) is the correct choice; substituting $\xi_v \text{dif_vap}$ would put the bracket at 6.36 instead of 0.133, predicting a shortfall of a factor of 58.

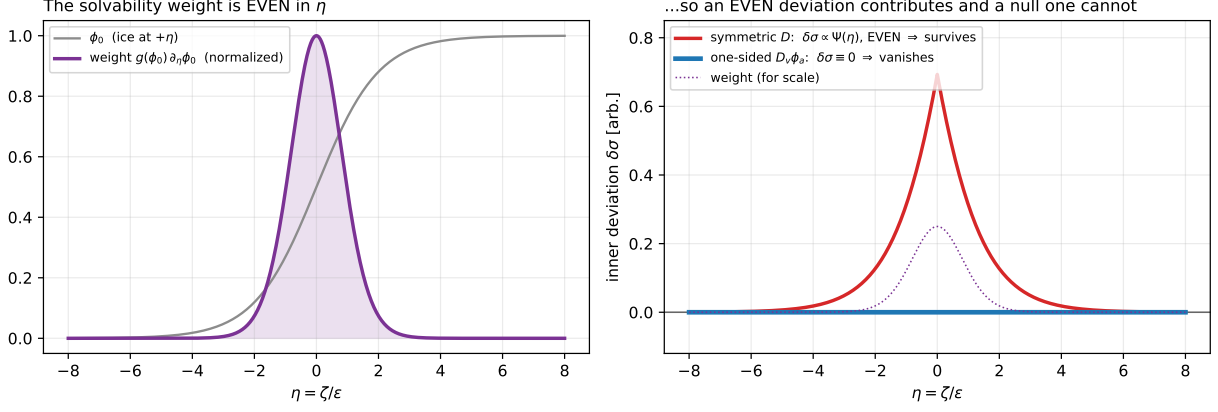


Figure 3: Structure underlying §5. Left: the solvability weight $g(\phi_0) \partial_\eta \phi_0$ is even in η and therefore samples only the even part of the inner deviation. Right: for symmetric D the deviation is even and non-vanishing, and the correction is retained; for the one-sided diffusivity $D_v \phi_a$ it vanishes identically, so that no correction arises for any weight.

6. Verification by parameter sweep

The preceding analysis rests on a single $(\epsilon, \beta_{\text{sub}0}, T)$ configuration. Because the offset is additive while $\beta_{\text{sub}0}$ is not, the predicted shortfall depends strongly on $\beta_{\text{sub}0}$, so that a sweep in $\beta_{\text{sub}0}$ discriminates between the candidate explanations without modification of the solver. Four runs were performed, identical except for `-beta_sub0`, each of 3406 steps and approximately two control-point spacings of meniscus displacement (batch 2026-08-25__18.14.15_beta_ladder):

$\times$	<code>-beta_sub0</code>	β_{bare}	predicted ratio	measured inner	measured outer	$\beta_{\text{fit}}/\beta_{\text{bare}}$
0.10	5.9216e4	1.8741e5	0.3160	0.3239	0.3082	1.0004
0.25	1.4804e5	2.7623e5	0.5359	0.5444	0.5254	1.0015
1.00	5.9216e5	7.2036e5	0.8220	0.8253	0.8126	1.0017
4.00	2.3686e6	2.4968e6	0.9487	0.9471	0.9429	1.0000

The measured ratio varies from 0.32 to 0.95, a factor of 2.9, and follows the predicted curve to within 2.3% in the worst case. The two alternative hypotheses are excluded by every arm: a constant value of 1.00, corresponding to the corrections operating as intended, and a constant value of 0.82, corresponding to a fixed factor of other origin.

The sweep varies only $\beta_{\text{sub}0}$ and carries a single vapor boundary condition per arm, so the two-parameter fit is ill-conditioned there; $d_{0,\text{sub}0}$ was measured to 0.06% on the nine-run 3×3 batch and is independent of β , and is therefore held fixed. This gives

$$\frac{\beta_{\text{fit}}}{\beta_{\text{bare}}} = 1.0009 \pm 0.0007 \quad \text{over a } 40 \times \text{ range in } \beta_{\text{sub}0}. \quad (13)$$

so that $C_{\text{corr}} = 0$ to better than one part in 10^3 . The correction implied by this result is given in §7.

7. Form of the required correction

7.1 Exact relation

The realized coefficient obeys, exactly and without further asymptotic analysis,

$$\beta_{\text{realized}} = \beta_{\text{sub0}} + \Delta, \quad \Delta \text{ from Eq. (6)}, \quad (14)$$

verified as $\beta_{\text{fit}}/\beta_{\text{bare}} = 1.0009 \pm 0.0007$ over a range of 40 in β_{sub0} . Eq. (14) is not a scaling estimate with a fitted $O(1)$ prefactor; it is a closed-form expression for a known spurious term, obtained from the model definition and confirmed to within 0.1 %.

7.2 Inadequacy of a fixed multiplicative factor

One might attempt to absorb the observed ratio 0.822 into β_{sub0} . This is not admissible, since 0.822 is not a constant: it equals $\beta_{\text{sub0}}/(\beta_{\text{sub0}} + \Delta)$, and Δ does not scale with β_{sub0} . Calibrating the factor at the $1.00\times$ arm, corresponding to $\beta_{\text{sub0}} = 0.7835\beta_{\text{target}}$, and applying it to the remaining arms gives

arm	$\Delta/\beta_{\text{target}}$	realized/target	error
0.10×	2.165	2.948	+195 %
0.25×	0.866	1.649	+65 %
1.00×	0.217	1.000	0 %
4.00×	0.054	0.838	−16 %

Applied to the Molaro configuration the same factor gives −4.4 % on the $D_v \times 10$ arm. The sense of such a factor is also ambiguous: $\Delta/\beta_{\text{target}} = 0.7835$ and $\beta_{\text{sub0}}/\beta_{\text{bare}} = 0.822$ are distinct quantities, and since $\beta_{\text{realized}} > \beta_{\text{sub0}}$ the model is over-resistive, so any compensating factor lies below unity rather than above it.

In one regime of interest no such factor exists. At the Molaro value $\epsilon = 1.2 \times 10^{-7}$ m with $\alpha_c \rightarrow 1$, $\beta_{\text{target}} = 7.94 \times 10^3$ while $\Delta = 1.79 \times 10^4$; Δ exceeds the target by a factor of 2.3, so pre-compensation would require a negative $-\beta_{\text{sub0}}$, which the option parser rejects. This is the attachment-limited regime relevant to a replication of Demmenie et al.

7.3 Equivalent formulations

The following two operations are algebraically identical:

- supply $\beta_{\text{sub0}} = \beta_{\text{target}} - \Delta$; or
- remove the two a_2 terms from τ_{sub} and supply β_{target} .

Both yield $\beta_{\text{realized}} = \beta_{\text{target}}$. The second is preferable on three grounds: it is applied once in the solver rather than in every run configuration; it cannot be invalidated by a subsequent change of ϵ or T ; and it remains well defined when $\Delta > \beta_{\text{target}}$, where pre-compensation admits no solution.

The required correction is therefore not a numerical factor but the removal of a counter-term for a contribution the model does not generate, with §5 providing the justification and §6 the verification.

8. Implementation

The change is to be made in `enceladus_DSM` and back-ported to `lunar_regolith_DSM`. The `enceladus` implementation is the more current of the two: it already uses the temperature-corrected D_v in the a_2 term, where the lunar code retains the base 0°C value, and its `assembly.c` documents the convention $W_{\text{KP}} = \sqrt{2}\epsilon$.

1. Make the correction explicit and optional in both the scalar path (`enceladus_main.c:852`, `lunar_main.c:829`) and the pointwise path (`SubKinetics`), which carry the same bracket:

```
PetscReal c_vap = a2 * eps / dv_T;
PetscReal c_therm = a2 * eps * drhovs_dT * lat_sub / k_eff;
if (!user.thin_iface_corr) { c_vap = 0.0; c_therm = 0.0; }
tau_sub = eps * lambda_sub * (beta_sub / a1 + c_therm + c_vap);
```

2. Set `-thin_iface_corr` to 0 by default, documenting at the flag the conditions under which it should be re-enabled: a two-sided vapor diffusivity, the addition of an anti-trapping current, or a regime in which $\epsilon v_n/D_v$ is not negligible. The flag is a model-structure switch and should not be used as a fitting parameter.
3. Correct the thermal coefficient in the enabled branch rather than restoring its present form. The derivative `drhovs_dT` is already computed and discarded at `lunar_main.c:735` ($8.1521 \times 10^{-5} \text{ kg m}^{-3} \text{ K}^{-1}$ at -20°C); `k_eff` is obtained from `ThermalCond` at $\phi = 1/2$.
4. Replace the base D_v in the lunar implementation with the temperature-corrected value, for consistency with `enceladus`.
5. Report $\beta_{\text{bare}}/\beta_{\text{sub0}}$ and the three bracket terms in the run header alongside τ_{sub} , so that each `outp.txt` records the separation between realized and requested kinetics.
6. Verify by repeating the $0.10\times$ arm of the parameter sweep (5.9 d simulated, 3406 steps), for which the predicted ratio changes from 0.32 to 1.00.
7. Revise the dependent calibration. `comp_eps.py` sizes ϵ using `beta_uns` in `eps_ceiling` and `kin_over_curv`; with the correction removed this relation ceases to be implicit. Re-derive ϵ for the standard -20°C geometries afterwards and confirm that the mesh element counts are unchanged.
8. Review previous results for dependence on absolute rates. Exponents and dimensionless ratios are unaffected; quantities such as times to a given state or absolute growth rates are displaced by the factor given in §4 for the corresponding (ϵ, T) .

9. Outstanding items

The spurious surface-diffusion contribution is the one $O(\epsilon)$ effect not constrained by the present work (§5). Resolving it requires either the one-sided asymptotic analysis for this model, by mapping onto Echebarria, Folch, Karma & Plapp, *Phys. Rev. E* **70**, 061604 (2004), whose weighting $q(\phi)$ corresponds to our ϕ_a , or the introduction of the anti-trapping current of Karma, *Phys. Rev. Lett.* **87**, 115701 (2001), which cancels it by construction:

$$\vec{j}_{\text{at}} = a_t \epsilon \rho_{\text{ice}} (\partial_t \phi) \frac{\nabla \phi}{|\nabla \phi|}, \quad (15)$$

added to the vapor flux, with a_t evaluated for the Moure & Fu well as a_1 and a_2 were. Neither is urgent, since the effects cancelled by the anti-trapping current scale as $\epsilon v_n/D_v$, here 2.7×10^{-14} .

Both should be completed before any conclusion that requires distinguishing numerical from physical surface transport.

The dominant discrepancy in the Molaro comparison is not kinetic. Removing the attachment resistance entirely increases the integrated neck-growth rate by a factor of 1.28 over the observed neck range, against a discrepancy of approximately 2; no adjustment to β accounts for it. The imposed far-field undersaturation is the larger influence: in arm B, $\sigma_\infty = -2.03 \times 10^{-3}$ against a curvature driving force at the neck of $+6.57 \times 10^{-4}$, larger by a factor of 3.1 and of opposite sign, so that the neck surface is in net sublimation. The A/B and dom2/dom3 comparisons already isolate this effect.